

# **PLC Calculator**

by

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## ABSTRACT

This project successfully implements a Prefix Expression Calculator that converts prefix expressions into infix notation, evaluates the result, and presents the output through a graphical user interface built using PyQt6. The system currently supports two arithmetic operations: addition and multiplication, and accepts both positive and negative integers as input.

The core functionality is powered by a single custom `PrefixParser` developed using the SLY (Simple Lex & Yacc) library. The `PrefixParser` handles both the evaluation of expressions and the construction of their corresponding infix representation. By managing these tasks within a unified parser, the system remains lightweight and efficient while maintaining clarity in its design.

The overall result is a functional and user-friendly calculator that highlights key concepts of parsing and GUI development. Future improvements may include expanding operator support, enhancing input validation, and adding more interactive features to the interface.

# CONTENTS

|                               | <b>Page</b> |
|-------------------------------|-------------|
| <b>ACKNOWLEDGEMENTS</b>       | <b>2</b>    |
| <b>ABSTRACT</b>               | <b>3</b>    |
| <b>CHAPTER 1 INTRODUCTION</b> | <b>5</b>    |
| <b>CHAPTER 2 METHODOLOGY</b>  | <b>6</b>    |
| 2.1 System Overview           | 6           |
| 2.2 Lexer Design              | 6           |
| 2.3 Parser Design             | 7           |
| 2.3.1 Grammar Rules           | 7           |
| 2.3.2 First and Follow Sets   | 7           |
| 2.3.3 Closure Table           | 7           |
| 2.3.4 SLR Parsing Table       | 9           |
| 2.4 GUI Design                | 10          |
| 2.4.1 GUI Components          | 10          |
| <b>CHAPTER 3 CONCLUSION</b>   | <b>11</b>   |

# CHAPTER 1

## INTRODUCTION

Mathematical expressions can be represented in three common notations: Infix, Prefix, and Postfix. Infix notation places the operator between operands (e.g.,  $2 + 3$ ), which is most familiar in standard arithmetic. Prefix notation (e.g.,  $+ 2 3$ ) places the operator before its operands, while Postfix notation (e.g.,  $2 3 +$ ) places the operator after the operands.

This project implements a calculator that:

- Accepts a prefix expression as input.
- Converts it into infix notation.
- Computes and returns the final result.
- Support positive and negative integers.
- Displays all outputs in a GUI using PyQt6.

The system is developed using Python and the SLY (Simple Lex & Yacc) library for parsing and evaluation, ensuring a structured and extensible parsing mechanism.

## **CHAPTER 2**

### **METHODOLOGY**

#### **2.1 System Overview**

The system is modular and comprises the following main components:

- MyLexer: Uses the SLY library to tokenize the input arithmetic expressions.
- PrefixParser: Converts prefix expressions to infix while performing evaluation.
- Graphical User Interface (GUI): Built with PyQt6 to support user interaction.

#### **2.2 Lexer Design**

The MyLexer class defines tokens for all supported operators, parentheses, numbers, and assignment symbols. To recognize integers (both positive and negative), the lexer uses the regular expression `@_(r'-?\d+')` for number tokenization.

This regular expression works as follows:

- `r' . . . '`: Defines a raw string in Python, treating backslashes as literal characters.
- `-?`: Matches an optional minus sign, allowing both positive and negative integers.
- `\d+`: Matches one or more digits, where `\d` represents digits from 0 to 9.

## 2.3 Parser Design

For this project, the chosen parsing technique is the SLR(1) parser, a type of bottom-up parser based on LR parsing with a simplified lookahead mechanism. This parser is suitable for handling arithmetic expressions written in prefix form.

### 2.3.1 Grammar Rules

The grammar used in this calculator project supports basic arithmetic operations in prefix notation. The rules are defined as follows:

- $S \rightarrow E$
- $E \rightarrow + E E$
- $E \rightarrow * E E$
- $E \rightarrow \text{number}$

### 2.3.2 First and Follow Sets

The computed FIRST and FOLLOW sets for the non-terminal symbols are:

| Nonterminal | FIRST            | FOLLOW       |
|-------------|------------------|--------------|
| S           | { +, *, number } | { \$ }       |
| E           | { +, *, number } | { \$, +, * } |

### 2.3.3 Closure Table

| State 0                             |
|-------------------------------------|
| $S \rightarrow \cdot E$             |
| $E \rightarrow \cdot + E E$         |
| $E \rightarrow \cdot * E E$         |
| $E \rightarrow \cdot \text{number}$ |

| State 1                 |
|-------------------------|
| $S \rightarrow E \cdot$ |

| State 2                             |
|-------------------------------------|
| $E \rightarrow + \cdot E E$         |
| $E \rightarrow \cdot + E E$         |
| $E \rightarrow \cdot * E E$         |
| $E \rightarrow \cdot \text{number}$ |

| State 3                             |
|-------------------------------------|
| $E \rightarrow * \cdot E E$         |
| $E \rightarrow \cdot + E E$         |
| $E \rightarrow \cdot * E E$         |
| $E \rightarrow \cdot \text{number}$ |

| State 4                             |
|-------------------------------------|
| $E \rightarrow \text{number} \cdot$ |

| State 5                             |
|-------------------------------------|
| $E \rightarrow + E \cdot E$         |
| $E \rightarrow \cdot + E E$         |
| $E \rightarrow \cdot * E E$         |
| $E \rightarrow \cdot \text{number}$ |

| State 6                             |
|-------------------------------------|
| $E \rightarrow * E \cdot E$         |
| $E \rightarrow \cdot + E E$         |
| $E \rightarrow \cdot * E E$         |
| $E \rightarrow \cdot \text{number}$ |

| State 7                     |
|-----------------------------|
| $E \rightarrow + E E \cdot$ |

| State 8                     |
|-----------------------------|
| $E \rightarrow * E E \cdot$ |



### 2.3.4 SLR Parsing Table

The SLR parsing table generated from the grammar is as follows:

| State | ACTION |    |        |     | GOTO |   |
|-------|--------|----|--------|-----|------|---|
|       | +      | *  | number | \$  | S    | E |
| 0     | s2     | s3 | s4     |     |      | 1 |
| 1     |        |    |        | acc |      |   |
| 2     | s2     | s3 | s4     |     |      | 5 |
| 3     | s2     | s3 | s4     |     |      | 6 |
| 4     | r3     | r3 | r3     | r3  |      |   |
| 5     | s2     | s3 | s4     |     |      | 7 |
| 6     | r1     | r1 | r1     | r1  |      |   |
| 7     | r2     | r2 | r2     | r2  |      |   |

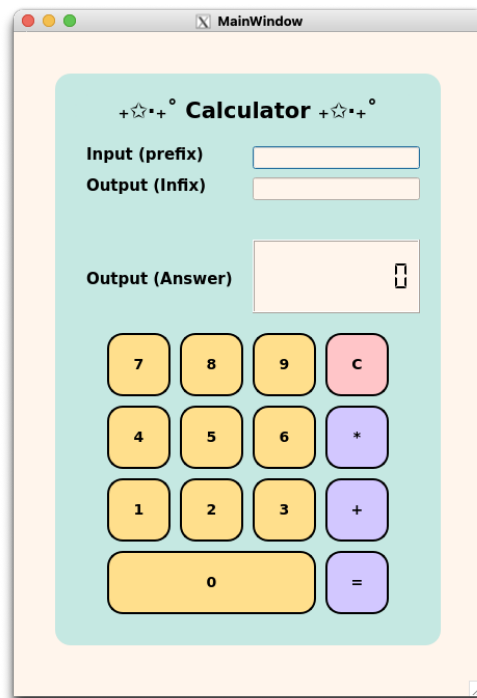
## 2.4 GUI Design

The GUI, built with PyQt6 and designed in Qt Designer, accepts prefix expressions and supports two operators: + and \*. It converts inputs to infix, evaluates the result, and displays both the infix form and result. The interface features a clean layout with color-coded buttons and hover effects for better interactivity.

### 2.4.1 GUI Components

The interface contains the following major components:

- **Input Field:** A QLineEdit named 'input\_prefix' for prefix input.
- **Infix Output:** A read-only QLineEdit named 'output\_infix' for infix output.
- **Answer Output:** A QLCDNumber widget named 'output\_answer' for results.
- **Digit Buttons:** Buttons 0–9, labeled 'button\_0' to 'button\_9' for integers.
- **Operator Buttons:** QPushButtons for '+', '\*' named 'button\_plus' and 'button\_star'.
- **Equal Button:** QPushButton labeled '=' named 'button\_equal'.
- **Clear Button:** QPushButton labeled 'C' named 'button\_clear'.
- **Labels:** QLabel widgets used to identify all input and output fields.
- **Container Frame:** A QFrame named 'frame' to organize the layout.



**Figure 2.1**

*Graphical User Interface of the Prefix Expression Calculator*

## **CHAPTER 3**

### **CONCLUSION**

This project successfully implements a Prefix Expression Calculator that converts prefix expressions into infix notation, evaluates the result, and presents the output through a graphical user interface built using PyQt6. The system currently supports two arithmetic operations: addition and multiplication, and accepts both positive and negative integers as input.

The core functionality is powered by a single custom `PrefixParser` developed using the SLY (Simple Lex & Yacc) library. The `PrefixParser` handles both the evaluation of expressions and the construction of their corresponding infix representation. By managing these tasks within a unified parser, the system remains lightweight and efficient while maintaining clarity in its design.

The overall result is a functional and user-friendly calculator that highlights key concepts of parsing and GUI development. Future improvements may include expanding operator support, enhancing input validation, and adding more interactive features to the interface.